

Title

Abstract— Customer preference analysis plays a critical role in a very competitive coffee business, but most businesses fail to effectively captures dynamic customer attributes and lifestyle factors. Today, coffee shops are highly dependent on generic beverage recommendation systems that operate on manual survey analysis and fixed customer segmentation models. These conventional techniques are very susceptible to subjective recall and they are not ideal in reflecting the fuzzy nature of changing consumer preferences. To eliminate these shortcomings, this paper conducts a literature review to suggest a data-driven recommendation architecture based on the e-commerce industry. Combining Natural Language Processing (NLP) specifically Word2Vec in semantic features extraction with Soft K-Means clustering, this study overcomes the binary nature of the other approaches. In the end, the approach would offer a systematic, evidence-based model of creating precise, customer-oriented beverage suggestions, which would improve menu planning and efficiency in operations.

Index Terms

Enter key words or phrases in alphabetical order, separated by commas.

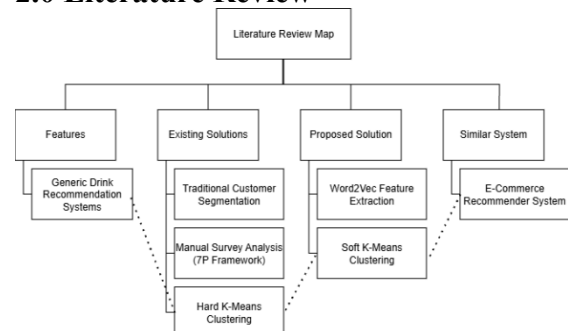
1.0 Introduction

Coffee industry is a highly competitive business environment whereby knowing the changing customer preferences is key in business sustainability. Yet, most coffee stores are not able to reflect on the effect of demographic and lifestyle factors on beverage preferences to correctly predict client preferences and the difference between perceived preferences and the reality of buying behaviour. But now, the decision-making processes are overly dependent on outdated forms of customer segmentation, like manual surveys using the 7P framework and strict analytical methods, such as the Hard K-Means clustering. There is too much subjectivity in such generic drink recommendation systems

and do not reflect the dynamic nature of consumer decision-making. As a result, these methods have a tendency to generate wrong products offerings and fail to generate the actionable and evidence based knowledge which already applied in the e-commerce industry successfully. To overcome this fundamental issue, it is necessary to perform systematic analysis of these constraints and investigate more advanced, data-driven analytical methods to be able to make more customer-specific recommendations.

In order to address these critical shortcomings, this paper will critically search the existing literature and outline a proposed methodology grounded on successful recommender systems. As a preliminary of this research, the next section provides a systematic Literature Review Map that serves as a roadmap of the study. This map gives a hierarchical view of the features and weaknesses of the generic drink recommendation systems, analyses the insufficiency of the solutions in the literature such as manual survey and hard clustering, and explains why the proposed solution integrates both Word2Vec and Soft K-Means clustering. In addition, it looks at a similar system in the e-commerce industry to confirm the availability of the proposed architecture. Based on this extensive literature review, the methodology section will further outline what the target users, sampling techniques, and data collection methods would be like to implement this framework practically.

2.0 Literature Review



In this section, critical review is done on the available literature about customer preference analysis and recommendation technologies. The structure of this review has been to formulate a Literature Review Map (see Appendix 1). The hierarchical relationship of the research components is visually presented in the form of the map, which is perfectly consistent with the formulated Research Questions (RQs).

It starts with an examination of the Features of generic drink recommendation systems that reveal the fundamental weaknesses experienced by coffee shops (RQ1). It then reviews Existing Solutions, where the author points to the flaws of the existing manual survey analysis and hard clustering procedures (RQ2). In order to fill these gaps, the review then reviews the Proposed Solution, which pays attention to the technical mechanisms of Word2Vec and Soft K-Means clustering (RQ3). Lastly, it explores one of the apposite Systems in the e-commerce business to support the cross-domain viability of the analytical method proposed. These domains will be discussed in the following subsections (2.1 to 2.4).

2.1 Feature

2.1 Features of Generic Drink Recommendation Systems

In order to find out the impact and vulnerability of generic drink recommendation systems, it is important to examine the core of the system and how it works in the coffee industries. This kind of systems usually has three main features such as data collection mechanisms, preference analysis, and the generation of recommendations.

2.1.1 Data Collection Mechanisms

The analytical process in generic recommendation systems is carried out by interpreting demographic and customer attribute data to learn the consumption patterns of target customers. This characteristic in the coffee industry strongly depends on structured, recollection-based questions using manual surveys, which in most cases rely on the classic marketing frameworks including the 7P model (Zamborita et al., 2024). Nevertheless, such a

mechanism of collect data creates a serious vulnerability. Since consumer buying behavior in coffee shops is usually affected by complicated psychological, social, and contextual factors (Mohd Ali et al., 2024), the use of retrospective surveys may result in subjective recall bias. The real emotional reactions and consumption pattern of customers are often hard to remember correctly, and this causes the input data entering the analysis to be inaccurate in nature.

2.1.2 Preference Analysis and Segmentation

The core processing feature of these systems involves analyzing the collected data to segment customers. Current industry practices frequently apply quantitative clustering algorithms or simple statistical tests to group customers based on observable purchasing behaviours and preferences (Zakaria & Mohd, 2024). Despite its widespread use, this analytical feature is highly vulnerable to the dynamic and shifting nature of consumer preferences (Safitri et al., 2025). Generic systems typically force customers into rigid, non-overlapping segments. This binary constraint fails to capture the "fuzzy" and ambiguous nature of human decision-making, where a single customer might prioritize both price perception and brand image simultaneously, rendering the resulting segments oversimplified and inaccurate.

2.1.3 Actionable Recommendation Generation

The final feature of generic systems is the generation of menu planning strategies or product suggestions based on the segmented data. While existing approaches may successfully identify general market trends or evaluate basic customer emotions in coffee shops (Mbougou et al., 2024), they exhibit a major vulnerability in translating these diagnostic insights into actionable, personalized product recommendations. This limitation exposes the decision-making process to misinterpretation, as store managers are forced to rely on generalized static reports rather than evidence-based, customer-specific drink matching, ultimately reducing the overall effectiveness of the recommendation system.

2.2 Existing solutions

2.2 Existing Solutions and Their Limitations

To answer RQ2, it is necessary to critically investigate the existing solutions currently adopted by the coffee industry for customer segmentation. While these traditional methods aim to interpret consumer behavior, literature reveals significant methodological limitations that hinder the accuracy of drink recommendations. The two most prominent existing solutions are manual survey analysis and hard clustering techniques.

2.2.1 Manual Survey Analysis and its Inefficiencies

A prevalent existing solution for gathering customer preference data is the use of manual surveys, frequently structured around marketing paradigms such as the 7P framework (Zamborita et al., 2024). While this approach provides a foundational understanding of consumer demographics, it presents severe operational shortcomings. According to Mohd Ali et al. (2024), relying on self-reported questionnaires introduces a high degree of subjective recall bias, as consumers often fail to accurately remember their psychological and social drivers during past purchases. Consequently, the data inputted into these systems is inherently flawed. Furthermore, the manual nature of data collection and subsequent processing creates significant operational latency. As Safitri et al. (2025) highlighted, the inability to process consumer feedback in real-time prevents businesses from swiftly adapting to dynamic market trends, rendering this existing solution inefficient for modern, fast-paced retail environments.

2.2.2 The Shortcomings of Hard K-Means Clustering

Once data is collected, existing systems commonly employ traditional statistical tools and Hard K-Means clustering to segment customers into actionable groups (Zakaria & Mohd, 2024). In this rigid analytical model, customers are forced into strict, non-overlapping categories, such as being exclusively classified as either "price-sensitive" or "brand-focused." However, literature indicates that this binary constraint is a major limitation. Consumer preferences in the

coffee sector are highly complex and often ambiguous, involving simultaneous considerations of taste, atmosphere, and price (Zamborita et al., 2024). Hard clustering techniques mathematically fail to capture this "fuzzy" and multifaceted consumer behaviour (Mbougou et al., 2024). By oversimplifying diverse customer profiles into rigid binary clusters, this existing solution generates inaccurate segmentations, which ultimately leads to generic and poorly targeted drink recommendations.

2.3 Proposed Solution

2.3 NLP and Soft Clustering

To address the critical limitations identified in existing customer-segmentation methods (RQ3), this research proposes the adoption of an advanced data-driven approach utilizing existing machine learning techniques. Specifically, the integration of Word2Vec feature extraction and Soft K-Means clustering offers a robust framework to overcome the inefficiencies of manual surveys and the rigidity of traditional clustering.

2.3.1 Word2Vec Feature Extraction for Unstructured Data

To resolve the subjective recall bias and high latency associated with manual surveys, the proposed solution replaces self-reported questionnaires with the automated ingestion of unstructured data, such as authentic customer reviews. However, raw text cannot be directly processed by clustering algorithms. Therefore, Natural Language Processing (NLP), specifically the Word2Vec technique, is employed to extract semantic features (Mbougou et al., 2024). By converting diverse and unstructured customer opinions into machine-readable numerical vectors, Word2Vec eliminates human recall bias and significantly reduces operational latency, providing a highly objective and efficient data foundation for preference analysis.

2.3.2 Soft K-Means Clustering for Fuzzy Segmentation

To overcome the binary constraints and oversimplification caused by Hard K-Means clustering, the proposed architecture incorporates the Soft K-Means algorithm.

Unlike traditional methods that force customers into single, rigid categories, Soft K-Means applies fuzzy logic, allowing a single data point (or customer profile) to belong to multiple sentiment or preference categories with varying degrees of probability (Mbougou et al., 2024). This probabilistic approach successfully captures the ambiguous and multifaceted nature of actual consumer decisions, thereby addressing the weaknesses of rigid segmentation tools and enabling highly accurate, customer-specific drink recommendations.

2.4 Similar system

To justify the feasibility and effectiveness of the proposed data-driven techniques for the coffee industry (RO3), it is essential to examine a similar system successfully implemented in a different industry. The e-commerce sector frequently utilizes analogous recommendation applications to analyze dynamic consumer behaviour and deliver personalized services. Table 1.0 below (Appendix 2) outlines a similar system utilized in the e-commerce industry, demonstrating how integrating NLP and sentiment analysis successfully drives recommendation engines.

The empirical study by Feng (2025) serves as a critical justification for this research's proposed methodology. Although operating in a different industry (E-commerce), the core application—processing unstructured customer feedback to generate personalized recommendations—is identical to the needs of the coffee industry. Feng's (2025) successful use of web information extraction and sentiment analysis demonstrates that transitioning away from manual surveys towards automated NLP pipelines effectively addresses data inaccuracy and operational latency. Consequently, adopting these existing techniques (Word2Vec and Soft K-Means) for coffee shop menu planning is a highly justifiable and empirically supported solution.

3.0 Methodology

3.1 Target user

To accurately evaluate the vulnerabilities of generic drink recommendation systems (RQ1), this research primarily focuses on the Direct Users of these systems: coffee shop managers, operational strategists, and menu planners. These individuals are responsible for analyzing customer data and are directly impacted by the high latency and rigid clustering of existing tools.

3.2 Sampling method

This research proposes the use of Purposive Sampling, a non-probability sampling technique, to select participants. This specific technique is chosen because it allows the researcher to deliberately select members of the population who possess specific operational experience in the coffee industry, ensuring highly relevant insights.

The selected sample will specifically target active coffee shop managers and menu planners operating in urban areas of Malaysia. The justification for this technique is strongly supported by existing successful research in the domain. For instance, Zamborita et al. (2024) successfully utilized purposive sampling to specifically target Millennials and Zoomers, proving that isolating a specific, highly relevant demographic yields more accurate preference insights. Similarly, by applying purposive sampling to selectively target experienced coffee shop managers, this research ensures that the data collected accurately reflects the true operational bottlenecks of current segmentation systems.

3.3 Data collection method

As this research is currently in the planning phase, the primary data collection will be executed using a single, highly focused quantitative approach, Online Survey Questionnaires.

The survey is selected as the optimal method because it allows for the structured and measurable evaluation of current operational challenges. The survey will be distributed to the purposively sampled coffee shop managers.

Drawing inspiration from the quantitative methodologies used by Safitri et al. (2025) to assess coffee sector preferences, this survey will be specifically designed to validate the shortcomings identified in the literature review (RQ2). The questionnaire will employ a 5-point Likert scale to ask managers about their experiences with subjective recall bias in current 7P framework surveys, the latency issues they face during manual data entry, and their perspective on adopting automated NLP pipelines like Word2Vec for menu planning.

4.0 Conclusion

In conclusion, this research proposal has systematically addressed the critical limitations of current customer preference analysis within the coffee industry. Through a structured literature review, the study highlighted the vulnerabilities of generic drink recommendation systems, particularly the subjective recall bias caused by manual survey analyses and the rigid, binary constraints of Hard K-Means clustering. To overcome these inefficiencies, this paper successfully proposed the adaptation of proven e-commerce technologies, specifically integrating Word2Vec feature extraction and Soft K-Means clustering. This proposed data-driven architecture is designed to process unstructured feedback and capture fuzzy consumer behaviours, directly addressing the identified literature gaps.

Furthermore, while this research is entirely in the planning phase and no actual development or data collection has been executed, the proposed methodology is strategically designed to efficiently support the research objectives. By utilizing purposive sampling to specifically target experienced coffee shop managers, the planned quantitative survey will efficiently gather highly relevant, empirical data regarding real-world operational bottlenecks. This planned investigation will ensure that the theoretical findings from the literature review are robustly supported by targeted, high-quality primary data. Ultimately, this planned framework aims to transition the coffee industry from assumption-based menu

planning to evidence-based strategies, paving the way for highly accurate and customer-specific drink recommendations.

5.0 References

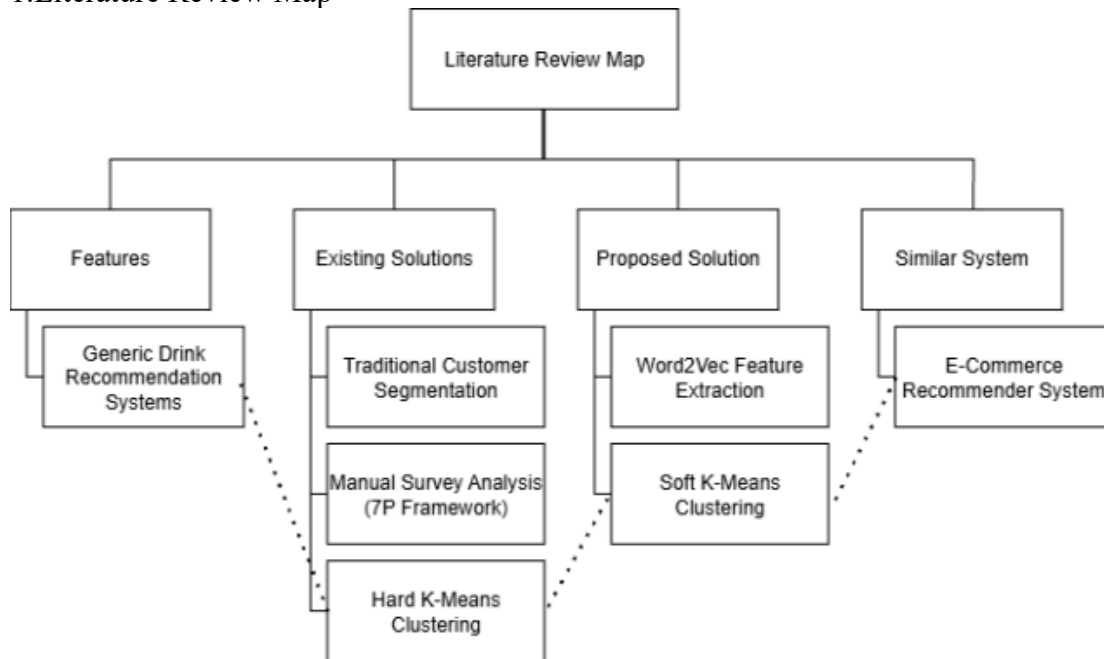
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Appendix

1.Literature Review Map



2. Table 1.0: Analysis of a Similar Recommendation System in the E-Commerce Industry

Attribute	Description
System/Article Title	E-commerce recommender system design based on web information extraction and sentiment analysis
Author(s) & Year	Feng, J. (2025)
Industry Domain	E-Commerce / Online Retail
Application Type	Product Recommendation Engine
Core Technologies Used	Natural Language Processing (NLP), Sentiment Analysis, Feature Extraction
Target Users	E-commerce consumers and online retailers
Data Collection / Sampling	Unstructured web information (Customer reviews and online feedback)
Key Results	Successfully integrated sentiment analysis with recommendation algorithms to provide highly personalized, evidence-based product suggestions, driving user satisfaction.
Limitations	Potential reliance on static web datasets which may require continuous updating to reflect real-time shifts in consumer trends.

3. Table 2.0: Questionnaire Design for Direct Users

No.	Question Description	Question Type	Response Format
Q1	"I frequently experience significant delays (latency) between collecting manual customer feedback and receiving actionable analysis reports."	Quantitative	5-Point Likert Scale (1 = Strongly Disagree to 5 = Strongly Agree)
Q2	"I believe that traditional customer surveys (e.g., questionnaires) often contain inaccurate data because customers struggle to perfectly recall their past experiences and true feelings (Subjective Recall Bias)."	Quantitative	5-Point Likert Scale (1 = Strongly Disagree to 5 = Strongly Agree)
Q3	"Current customer segmentation methods force customers into rigid, 'black-and-white' categories (e.g., strictly 'price-sensitive' or 'brand-focused'), which fail to capture their complex and overlapping taste preferences."	Quantitative	5-Point Likert Scale (1 = Strongly Disagree to 5 = Strongly Agree)
Q4	"My coffee shop struggles to quickly adapt menu offerings to shifting market trends due to a lack of automated, real-time data analysis."	Quantitative	5-Point Likert Scale (1 = Strongly Disagree to 5 = Strongly Agree)
Q5	"Implementing an automated system that analyzes unstructured customer reviews (using Natural Language Processing) and allows for flexible, overlapping customer profiles would significantly improve my menu planning decisions."	Quantitative	5-Point Likert Scale (1 = Strongly Disagree to 5 = Strongly Agree)
Q6	What is the most significant operational challenge you currently face when trying to interpret your customers' shifting preferences using your existing methods?	Qualitative (Open-ended)	Text Box / Short Essay
Q7	If an automated, AI-driven recommendation system were implemented in your coffee shop, what specific insights or features would be the most valuable for your daily menu planning and promotional strategies?	Qualitative (Open-ended)	Text Box / Short Essay